

Development of near-surface perched water tables during natural and artificial stemflow generation by babassu palms

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SUMMARY

Rainfall falling on canopies is redistributed and reaches the soil surface as throughfall or stemflow. I investigate stemflow generation by babassu palms (*Attalea speciosa* Mart.) in a Brazilian agroforestry system and its fate once the stemflow water reached the soil surface. Rainfall, stemflow and perched water tables were monitored on rainfall-event basis. Dye tracer experiments monitored stemflow-induced preferential flow paths. Root distributions were related to soil water redistribution. Perched water tables and roots were studied only on adult palms, while stemflow measurements and dye tracer experiments were performed on young palms, too. Average rainfall-collecting area per adult palm was $6.4 \pm 0.3 \text{ m}^2$ ($\pm \text{SE}$). Funneling ratios (ratio of stemflow volume and basal area normalized by rainfall depth) ranged between 16–71 and 4–55 for adult and young palms, respectively. On average, $9.1 \pm 3.1 \text{ L}$ ($\pm \text{SE}$) of adult palm stemflow were intercepted. For adult palms, stemflow induced soil saturation and, hence, perched water tables were detected more frequently near adult palm stems than further away. Tracer experiments at adult palms revealed initial preferential horizontal flow, with subsequent downward water movement towards the lower wetting front. Meanwhile, young palms funneled rainfall via their fronds directly to their subterranean stems. Coarse roots density of adult palms was highest next to the subterranean stem and, consequently, palms might profit from increased water input compared to open area rainfall. Beside this positive effect for this very abundant palm species in Brazil, the generation of soil saturation next to the palm stems might also lead to negative environmental impacts. As discussed soil saturation and simultaneous input of nitrate and presumably dissolved organic carbon could lead to hotspots of denitrification and, hence, greenhouse gas emissions.

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1. Introduction

As rainfall passes through canopies water is redistributed spatially and reaches the soil as throughfall and stemflow. Stemflow as a fraction of incident rainfall varies widely across climatic regions and vegetation type due to differences in meteorological conditions, stand densities or the three dimensional geometry of canopy structure (Levia and Frost, 2003). For studies reviewed by Levia and Frost (2003), maximum reported stemflow as a percentage of incident rainfall were 13.6%, 20.0%, and 45.0% for tropical, temperate, and semi-arid and arid regions, respectively. Plants with funnel-like canopy shapes can produce high stemflow volumes that can lead to more than 20% stemflow of gross rainfall in monocultures (Cattan et al., 2007, data not shown; Schroth et al., 1999). These include babassu (*Attalea speciosa* Mart.) and oil palms (*Elaeis guineensis* Jacq.), which grow in forests as well as in plantations or agroforestry systems (Banabas et al., 2008; Germer et al., 2010b). The redistribution of rainfall by the canopy

leads to an uneven water input to the soil surface that might influence groundwater recharge or the development of spatial soil moisture patterns (Levia and Frost, 2003). In addition, stemflow is usually enriched in nutrients compared to rainfall, and stemflow often has even higher solute concentrations than throughfall (Levia and Frost, 2003; Parker, 1983). For instance, stemflow from babassu palms compared to rainfall can be enriched 81-fold in nitrate and twofold in ammonium (Germer et al., 2012). This can lead to localized input to the soil of nitrogen, which is often limiting in crop production.

It has been shown that highly localized water input by stemflow can lead to Horton overland flow (Banabas et al., 2008; Cattan et al., 2007; Herwitz, 1986). Over the last two decades, evidence of stemflow-induced higher soil moisture below trees or shrubs compared to soil further away from them has been reported for both semi-arid and very wet environmental conditions (Cattan et al., 2007; Durocher, 1990; Gonzalez-Hidalgo and Bellot, 1997; Ladekarl, 1998; Li et al., 2008; Liang et al., 2011; Ludwig et al., 2005; Návar, 2011; Sansoulet et al., 2008). Consequently, stemflow can, for instance, contribute up to 20% of total groundwater recharge for a pine forest while making up just 0.7% of total rainfall

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(Tanaka et al., 1996; Taniguchi et al., 1996) or induce soil saturation at the soil–bedrock interface, just below individual trees such as reported for red oak (*Quercus rubra* L.) in England or tall stewartia (*Stewartia monadelphica* Sieb. & Zucc.) in Japan (Durocher, 1990; Liang et al., 2011; Liang et al., 2009). Tracer studies showed that quick, deep percolation can be enhanced by preferential flow along the coarse roots of tall stewartia and beech trees in temperate climates (Liang et al., 2011, 2009; Schwärzel et al., 2012) as well as desert shrubs (Li et al., 2009; Martinez-Meza and Whitford, 1996) on soils without impeding layers. With increasing amount of rainfall per event, stemflow quantity increases and, hence, preferential flow can reach deeper soil layers (Li et al., 2009).

It can, however, be expected that soil texture has a substantial influence on water movement. Babassu palms grow predominantly in soils with a clay accumulation layer (such as Ultisol) and they can generate great stemflow volumes due to their funnel-like canopy. To my knowledge, no study has investigated the fate of nutrient-rich palm stemflow that reaches the ground. A crucial difference between palms and other kinds of trees is that palm roots lack secondary growth (Hodel, 2009), which was assumed to facilitate the generation of preferential flow along roots (Johnson and Lehmann, 2006). These soil textural and root growth characteristics combined with high stemflow fluxes by babassu palms might rather result in the establishment of near-surface perched water tables than quick deep percolation. Perched water tables can establish in Ultisols under pasture as well as under forest (Germer et al., 2010a). In addition, the generation of soil saturation by stemflow from babassu palms enriched in nitrate might lead to hotspots of denitrification next to the base of the palms and, hence, to greenhouse gas emissions.

This potentially negative environmental effect might be of particular relevance as the babassu palm is abundant and has high socioeconomic importance in South America. In Brazil it grows on 18.4 million hectares (Pinheiro, 2004) and only in the Maranhão State secondary forests of babassu palms occupy 10.3 million hectares (Pinheiro, 2004). The omnipresence of the babassu palm in pastures and cropland led to traditional agroforestry systems: agrosilviculture (trees with crops) and silvipasture (trees with pasture) (Kass et al., 1993). The babassu palm with its oil-rich fruits has a high economic potential not only for biofuel production. Several subsistence products as charcoal, animal feed, oil, construction materials are made from babassu palms (Pinheiro, 2004).

In order to study stemflow generation from babassu palms and the fate of the stemflow once it reached the soil, the objectives of this study were to: (1) quantify stemflow and the amount of rainfall intercepted by the palm trunk without reaching the soil (trunk storage) of isolated babassu palms, (2) relate rainfall intensity to observed stemflow amounts, (3) compare the frequency of perched water table generation at different distances from palm stems, (4) characterize water percolation in the stem-near soil zone, and (5) describe the root-length density distribution of adult babassu palms.

2. Study area and methods

2.1. Study area

The study was carried out in the *Zona dos Cocais* in northeastern Brazil. The study site was located next to the Itapiracó environmental-protection area in the city of São Luís, Maranhão State/Brazil (2° 31' 36" S, 44° 12' 24" W). The climate is classified as tropical wet and dry (Köppen's Aw, Peel et al., 2007). For the 25 years from 1987 to 2011, average annual temperature was 27.7 ± 0.4 °C (mean $\pm$ 1 SD, range: 26.6–28.4 °C), and average annual precipitation was 2200 ± 495 mm (mean $\pm$ 1 SD, range:

1198–2846 mm), with a pronounced dry season from August to November of less than 60 mm of mean monthly rainfall (Fig. 1, INMET, 2012).

Field work was conducted on three adjacent plots with no obvious inclination (Fig. 2). Plot 1 (size: 1500 m²) was fallow land with isolated adult babassu palms intergrown with a grass and shrub cover, grazed by one horse prior to this study. Plot 2 (size: 750 m²) was a secondary forest dominated by babassu palms and grazed by five sheep and plot 3 (size: 750 m²) a mixed secondary forest regrowth area, without grazing, situated at the edge of the Itapiracó protection area. The soil for all three plots, developed from the Itapicuru sandstone formation, is a sandy loam overlain by loamy sand (Table 1).

2.2. Field measurements

Rainfall was sampled manually with a standard rain gauge (22.5 cm diameter) in an open area along the margin of plot 1. In addition, rainfall intensities, sums and timing were obtained from pluviographs, recorded at a weather station within 600 m of the study area and operated by the National Meteorological Institute (INMET). Rainfall events were separated by at least 5 h without rainfall to ensure that stemflow from previous events stopped before.

Morphologic characteristics of all monitored palms are summarized in Table 2. For stemflow measurements, all eight adult palms present on plot 1 and three young palms with short subterranean stems of different sizes on plot 2 – for which sheep grazing damage was minimal – were equipped with stemflow collectors. The stemflow collectors for the adult palms were made of aluminum, with an inner diameter of 5 cm, and fitted around trunks at breast height (1.3 m above the soil surface) in a downward spiral (Germer et al., 2010b). Polythene foil inlays were sealed to the trunks with silicone glue. The same polythene film was attached in a downward spiral to young palm fronds with thumbtacks and sealed. Funnels were attached to the lower end of collectors, and flexible tubes diverted the water to canisters standing on the soil surface. Collectors were cleaned of litter to avoid clogging and checked for leakage daily. Stemflow monitoring started on March 28,

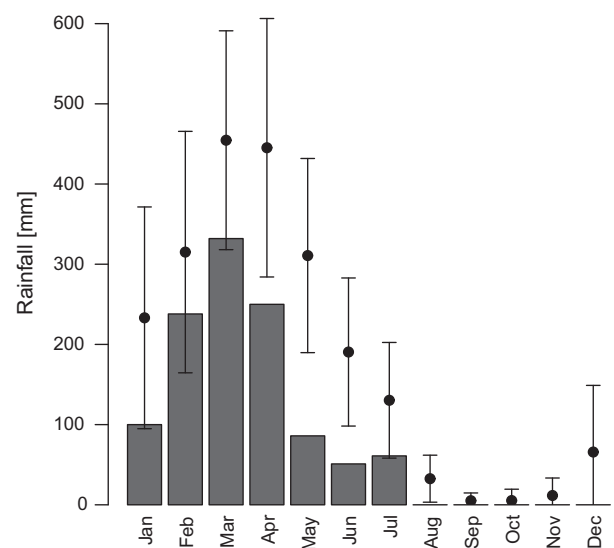


Fig. 1. Monthly rainfall for January through July 2012 (bars). The solid circles and the vertical lines are the mean and standard errors, respectively, for the period 1987–2011 (data source: INMET, 2012).

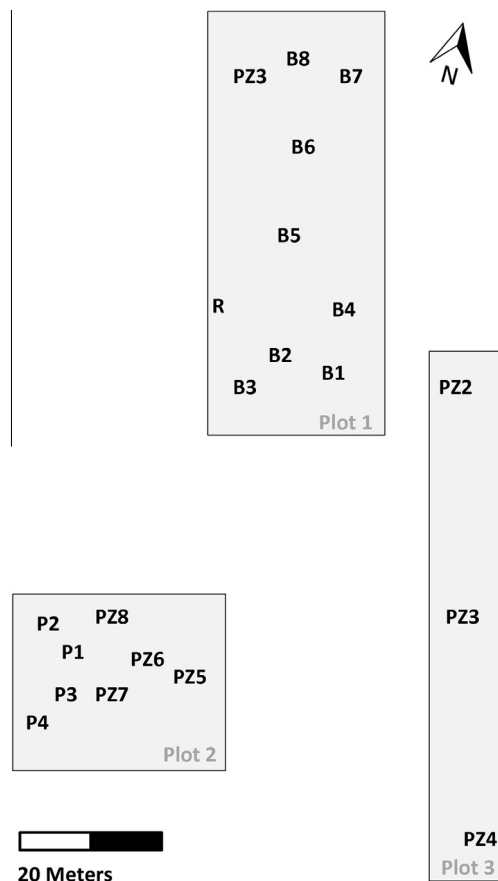


Fig. 2. Sketch of the approximate positions of plots and trees selected for this study. Adult (B) and young (P) babassu palms were monitored for stemflow generation and babassu palms of varying size for perched water table monitoring (PZ). Rainfall (R) was sampled in Plot 1.

Table 1

Percentage of sand, silt, and clay as well as soil texture class per sampled soil depth of Plot 2 ($n = 2$ per depth).

Soil depth (cm)	Sand (%)	Silt (%)	Clay (%)	Soil texture class
0–30	83	12	5	Loamy sand
30–80	62	21	17	Sandy loam
80–100	67	20	13	Sandy loam

2012 for the adult palms and on April 09, 2012 for the young ones, ending for both on July 04, 2012.

Thirty piezometers (PVC tubes with 5 cm diameter) were installed to monitor the occurrence of perched water tables at soil depths of 20 and 50 cm and at different distances from each palm tree. Palms with piezometers were distributed over all tree plots (Table 2). According to the palm canopy dimension, the first distance from palm stem (20 cm) was assumed to represent an area of direct stemflow influence; the second distance (100 cm) is an area of lower water input, due to shading from rainfall by palm fronds; meanwhile, the third distance (200 cm) corresponds to an area of intermediate water input in relatively open areas. As sites PZ5 to PZ8 were located in the secondary forest, no piezometers were installed at the 200 cm distance for them. To record maximum water levels in the piezometers, self-made indicator tubes were inserted into them and read daily (crest stage gauge). These were made of transparent plastic tubes that had been perforated at 1 cm intervals, closed at the bottom end with a cap and filled with enough cork crumbs to cover the bottom. Rising water in

Table 2

Morphologic palm characteristics for adult (B) and young (P) babassu palms monitored for stemflow generation and babassu palms of varying size for perched water table monitoring (PZ). The trunk diameter at breast height (DBH) is given only for those palms whose stems were not covered with palm fronds at a height of 1.3 m above soil surface. The plot number indicates in which plot each palm was growing.

Palm	DBH (cm)	Trunk height (m)	Number of palm fronds	Plot number
B1	37.2	6	20	1
B2	35.0	6	15	1
B3	35.0	7	12	1
B4	33.4	6	18	1
B5	36.9	6	22	1
B6	29.3	6	16	1
B7	38.2	5	21	1
B8	35.3	6	18	1
P1	–	–	10	2
P2	–	–	6	2
P3	–	–	6	2
PZ1	39.8	6	19	3
PZ2	30.9	7	15	3
PZ3	–	3	14	3
PZ4	–	0	7	3
PZ5	30.2	8	15	2
PZ6	32.5	6	15	2
PZ7	–	5	12	2
PZ8	–	4	11	2

the tubes forced the cork to rise, and some cork crumbs remained stuck to the tube at maximum level, even after water level recession. Hence, maximum water level could be registered. After each event the cork crumbs were pushed down with a thin stick. Only water levels above 2 cm could be detected, however, as this was the lower end of tube perforation. The indicator tubes were tested with tap water before installation. Piezometer observations were performed from April 11 to July 04, 2012.

Dye-tracer experiments were carried out to visualize stemflow water percolation through the soil next to two adult palm trunks (PZ5 and PZ6) and one young palm (P3). For each experiment, the tracer Brilliant Blue was mixed at a concentration of 4 g L^{-1} with water in a canister, standing on a support 1.5 m above the soil surface. For stemflow simulation of a 10 mm rainfall event, the needed water volume was estimated by using stemflow quantities measured from mid April to mid May. For adult and young palms 52 L and 9 L of water were used, respectively. The canister was equipped with a tap at its bottom edge. Flexible tubing was connected to the tap to divert the water to the palm trunks of adult palms and palm fronds of the young palm 20 cm above the soil surface. Along the trunk or palm frond circumference, the flexible tubing was perforated at 5 cm intervals to allow water to flow onto the trunk. Decreasing water pressure due to decreasing water level in the canister was compensated manually with a flow-rate regulator (10 to 100 L h^{-1} , PKP Process Instruments Inc.). A flow rate of 80 L h^{-1} was applied to the first adult palm (PZ5) for 20 min, followed by 40 L h^{-1} for an additional 40 min representing rainfall intensities of 15 mm h^{-1} and 8 mm h^{-1} , respectively. No data on rainfall intensities were available at this point of the study. But as occasional rainfall depth observations during rainfall events suggested that short term rainfall intensities can be higher, flow rates were increased and flow duration decreased for the second adult palm (PZ 6), with 100 L h^{-1} for 20 min followed by 80 L h^{-1} for 14 min simulating rainfall intensities of 19 mm h^{-1} and 15 mm h^{-1} , respectively. As the water volume for the young palm stemflow simulation was only 9 L, the minimum flow rate of the regulator of 10 L h^{-1} was kept for 54 min. For PZ5 palm, a stemflow simulation test run was performed. During the test run and both tracer experiments, piezometric water levels were recorded every 5 min by checking the wetting height on the indicator tubes. The

indicator tubes were dried with a paper towel after each measurement. Soil profiles were excavated the next day, and dye patterns were photographed with a digital camera. Soil profiles extended from 0.2 to 2.2 m from each palm trunk, with 0.7 m width and 1.0 m depth, and formed a narrow groove up to the subterranean stem.

Root length density (RLD) was studied at the two soil profiles next to the adult palms according to the profile-wall method described by Böhm (1979). The working face of the profile was first smoothed with a machete. All roots sticking out from the profile wall were then cut. To facilitate root counting, a grid net of 1 m² was placed against the profile wall. One after the other, 10 cm² grid cells, with a 5 mm thick soil layer, were removed with a screw driver, and all babassu root length units of fine (<2 mm) and coarse (≥2 mm) roots were counted separately. One root unit was assumed to equal the soil layer depth that had been previously removed (5 mm).

2.3. Data analysis

Only rainfall events sampled on an event basis were used for all statistical analysis. Spearman rank correlation coefficients were used to evaluate potential relations between (a) median stemflow of adult palms and rainfall per event and (b) stemflow of individual adult babassu palms and DBH or number of palm fronds.

Visual examination of scatterplots of median stemflow volumes per event against rainfall volumes for differing rainfall intensity classes suggested that stemflow as a fraction of incident rainfall decreases with increasing rainfall intensity (Fig. 3). In order to analyze and discuss stemflow parameters (e.g. trunk storage), further statistical analysis was done separately for low- and high-intensity rainfall events, with a maximum 10-min rainfall intensity per event below 30 mm h⁻¹ and between 30 and 60 mm h⁻¹, respectively. To compare slopes, an analysis of covariance (ANCOVA) was performed for stemflow volume on rainfall depth separately for low- and high-intensity rainfall events. Normality and homoscedasticity were checked with a Shapiro–Wilk test (Shapiro and Wilk, 1965) on residuals and residual plots. As residuals were not normally distributed, the ANCOVA was repeated with ranked data (Conover and Iman, 1982).

The stemflow funneling ratio (FR) is the stemflow volume per collecting area, normalized by rainfall depth [unitless]. According to Herwitz (1986), FR is defined as

$$FR = S / (B * R) \quad (1)$$

where S is stemflow (L), B is the palm stem basal area (mm²), and R the rainfall depth (mm). A funneling ratio below one indicates that less water flows down the stem than open-area rainfall over an equivalent area, and a funneling ratio exceeding one indicates that the crown of the plant funnels water towards its stem. The funneling ratio for the study period per palm was calculated as total stemflow yield normalized by total rainfall depth during the study period and then divided by the tree or palm basal area.

For all data analysis, I used the language and environment of R (Version 2.10.0, R Development Core Team, 2009).

3. Results

3.1. Stemflow quantity and parameters

During the study period stemflow was measured 44 times, including 35 individual rainfall events. The remaining nine measurements included more than one rainfall event. Total rainfall over the study period was 455 mm. The 35 rainfall events sampled on an event basis had a total of 330 mm rainfall. But it turned out that monthly rainfall sums during 2012 were much lower than 25-year averages (Fig. 1). Rainfall amounts of the 35 individual rainfall events ranged from 0.3 to 36.3 mm. Median maximum 10-min and 60-min rainfall intensities were 15.6 ± 18.7 mm h⁻¹ ($\pm$ median absolute deviation from the median (MAD)) and 3.8 ± 4.4 mm h⁻¹ ($\pm$ MAD) and ranged from 1.8 to 90.0 mm h⁻¹ and 0.4 to 28.2 mm h⁻¹, respectively. No event went on for longer than 9 h and half of the events lasted less than one hour. Stemflow was generated by events as small as 1 mm, but when palm stems were still wet, 0.25 mm of rainfall already triggered stemflow.

In order to quantify stemflow generation and trunk storage per stand, calculations were done with median stemflow volume per event. Results are only reported for the eight adult palms. During the first few millimeters of rainfall, the trunk storage of the adult palms became filled up (intercept with rainfall axis in Fig. 3). For rainfall events greater than 2.5 mm, median adult palm stemflow volume and rainfall depth per event had a significant and strong positive correlation of 0.97 and 0.95 for events with low (below 30 mm h⁻¹) and high (between 30 and 60 mm h⁻¹) maximum 10-min rainfall intensities, respectively (Spearman correlation, $p < 0.05$). The stemflow regression for low-intensity rainfall events with a rainfall depth >2.5 mm is given by:

$$S = 6.4(\pm 0.3)R - 9.1(\pm 3.1), (\pm SE), R^2 = 0.98, p < 0.001, n = 11. \quad (2)$$

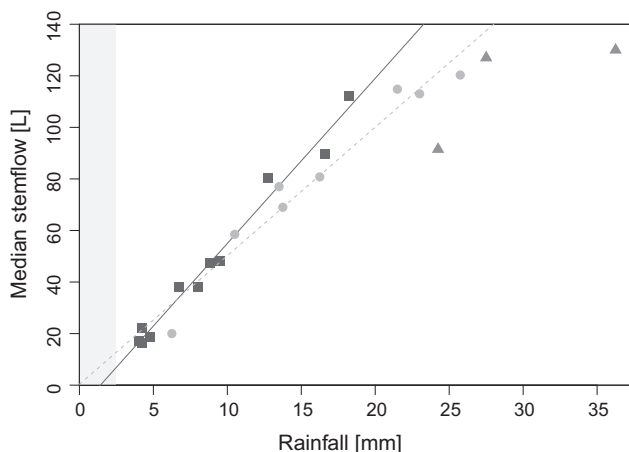


Fig. 3. Relationship between median stemflow volume of adult babassu palms and rainfall depth for rainfall events with low (circles, ≤ 30 mm h⁻¹), high (squares, >30 and ≤ 60 mm h⁻¹) and extremely high (triangles, >60 mm h⁻¹) maximal 10-min rainfall intensities. Regression lines for all events > 5 mm of rainfall are plotted separately for low (solid) and high (dashed) intensity rainfall events.

The regression slope of median stemflow volume per event on rainfall depth represents the volume of stemflow per millimeter of rainfall, which equals the vertical projection of the stemflow-contributing vegetation surface area, because 1 mm rainfall on one square meter results in one liter of water. Hence the regression slope of Eq. (2) indicates a stemflow-contributing area of 6.4 ± 0.3 m² ($\pm$ SE) assuming that no splash loss from that area took place for the low-intensity rainfall events. For high-intensity rainfall events, the slope of 5.0 ± 0.5 ($\pm$ SE) tended to be lower than that of low-intensity rainfall events (Fig. 3, no significant differences according to ANCOVA with ranked data, $p < 0.05$). A lower slope for high-intensity compared to low-intensity rainfall events would imply that the stemflow-contributing area was less effective. For median stemflow volume per event regressed on rainfall depth for events >2.5 mm and with low rainfall intensity (<30 mm h⁻¹), the absolute value of the intercept with the stemflow axis is considered to equal trunk storage capacity (Lloyd et al., 1988). Small rainfall events were excluded, because during the first few

millimeters of rainfall the trunk became wet and, hence, the relationship between rainfall and stemflow was curvilinear and high intensity events were excluded due to possible splash loss. The trunk storage calculated for median stemflow volume per event was 9.1 ± 3.1 L ($\pm$ SE) of rainfall for adult palms. When including only the events with antecedent dry periods (no rainfall) greater than 24 h, then the trunk storage increased to 11.0 ± 3.8 L ($\pm$ SE) of rainfall.

In order to assess the variability of stemflow generation and trunk storage for different palms, stemflow volume of individual palms was analyzed. For all individual young and adult palms, the regression slope of stemflow volume per palm and event on rainfall depth tended to be lower for high- than for low-intensity rainfall events (Table 3; again, no significant differences according to ANCOVA with ranked data, $p < 0.05$). The trunk storage tended to be higher for adult palms when excluding events with antecedent dry periods shorter than 24 h (Table 3). For low-intensity rainfall events, the slope or contributing area of adult palms showed a weak but significant correlation with DBH (0.4, $p < 0.05$) and with the number of fronds per palm (0.6, $p < 0.1$). No significant correlation was found for trunk storage and DBH or the number of palm fronds.

The median funneling ratios for rainfall and stemflow totals of the whole study period was 50 ± 10 (median ± 1 MAD) for the eight adult palms, indicating that on half of the palms increased at least 50-fold the water input to the bases of their stems, compared to open-area rainfall. For individual adult palms the funneling ratio for the whole study period ranged from 16 to 71, whereas for the three young palms it was 4, 7, and 55. Funneling ratios of individual palms changed with event size and intensity (Fig. 4). In general funneling ratios tended to increase during the first millimeters of rainfall. For rainfall events greater 10 mm, the variability of funneling ratios per event could be low (B5, range: 40–65) or high (B8, range: 20–100). For maximum 10-min rainfall intensities greater than 10 mm h^{-1} , funneling ratios tended to decrease with increasing rainfall intensity. For young palms, patterns were less obvious (Fig. 4).

3.2. Perched water table detection

Hydraulic conductivity can decrease relatively abruptly with increasing soil depth. Above an impeding layer with relatively low soil hydraulic conductivity percolating water can accumulate and form a perched water table. The piezometer readings represent maximum levels of perched water tables established during rainfall events. For half of the monitored palms, I more frequently detected shallow perched water tables close to their stems than further away from them (Table 4). A 3.9 mm rainfall event on June

23 was the smallest rainfall event for which a perched water table was detected (PZ1, Fig. 5). The event with the lowest maximal 10-minute rainfall intensity that triggered a piezometer reaction was registered on April 11, with 6.0 mm h^{-1} (PZ1). The highest perched water levels detected in 20 cm soil depth were 18 cm in PZ5 on April 26 and May 12 with maximum 10-min rainfall intensities of 31.2 mm h^{-1} and 90.0 mm h^{-1} , respectively. The majority of all water level records were above 5 cm (Fig. 5). Water level detection in 50 cm soil depth was rare and never exceeded 4 cm.

Perched water table establishment was also monitored during two 1-h stemflow simulations of a 10 mm rainfall event. Among 40 events monitored for perched water tables 12 had rainfall amounts greater than 10 mm. The maximum 60-min rainfall intensities for these 12 events were $\geq 8 \text{ mm h}^{-1}$, but maximum 10-min rainfall intensities were always 3- to 6-fold higher. Perched water table establishment during simulated stemflow was only registered in the near-stem piezometer (distance: 20 cm) and at shallow soil depth (20 cm) for PZ5. Increasing water levels for both runs (test and tracer experiment) were first detected after 10 min of stemflow, with a stemflow intensity of 80 L h^{-1} representing a rainfall intensity of 15 mm h^{-1} , which was reached by half of the monitored events (Fig. 6). The reduction of stemflow intensity to 40 L h^{-1} and the cessation of stemflow led to an immediate drop of the water levels. Even though stemflow intensity was the same for both runs, the temporal patterns of their water levels were quite different (Fig. 6). Water levels increased more quickly, reached a higher level, and then decreased more quickly during the first run. The perched water table extinguished 20 and 60 min after stemflow ceased for the first and second runs, respectively. In the 50 cm deep piezometer, I observed water 75 min after the onset of stemflow during the first event, but the water level remained below 2 cm and, hence, no piezometer response was recorded. During the second run, no water was observed in this piezometer.

3.3. Tracer experiment

For the tracer experiments, all artificially applied stemflow water infiltrated into the soil and no overland developed. For the adult palms, the dye tracer patterns reveal horizontal preferential flow in the shallow subsoil. For PZ5, the dye tracer was visible up to a horizontal distance of 25 cm from the palm stem at a depth of 10 cm (Fig. 7a). As far as 46 cm from the stem and at 12 cm soil depth, a root covered with blue dye also indicated occasional preferential flow along horizontal roots. For PZ6, the dye tracer extended up to 40 cm from the palm stem at a depth of 7 cm. The young palm, by contrast, directed the rainfall over its fronds directly to its subterranean stem and roots at 25–50 cm soil depths

Table 3
Stemflow parameters trunk storage (intercept with x-axis) and stemflow contributing area (slope) for low-intensity events, trunk storage for low-intensity events that had an antecedent dry period (ADP) of at least 24 h, and the contributing area for high-intensity events. The stemflow parameters were derived by linear regression models of stemflow volume (S) on rainfall depth (R) per adult palm (B1 to B8) or young palm (P1 and P3; not P2, as no significant regression equation was found). The regression equation is given by $S = \text{slope} \cdot R + \text{intercept}$ whereas the slope is interpreted as the contributing area and the absolute value of the intercept as the trunk storage. The stemflow parameters were calculated separately for rainfall events with low ($\leq 30 \text{ mm h}^{-1}$) and high (> 30 and $\leq 60 \text{ mm h}^{-1}$) maximal 10-min rainfall intensities.

	Low-intensity events			Low-intensity events, ADP > 24 h		High-intensity events	
	Trunk storage (L)	Contributing area (m^2)	R^2	Trunk storage (L)	R^2	Contributing area (m^2)	R^2
B1	18.5	11.3	0.97	23.6	0.98	8.3	0.97
B2	14.0	6.3	0.97	17.3	0.98	5.3	0.97
B3	12.9	3.0	0.90	15.0	0.91	1.6 [†]	0.74
B4	8.8	6.1	0.97	12.8	0.98	3.9 [†]	0.68
B5	13.6	7.4	0.99	14.3	0.99	4.3	0.93
B6	12.7	5.7	0.93	17.9	0.95	3.9	0.99
B7	3.9	7.1	0.92	10.8	0.96	4.8	0.94
B8	20.8	10.5	0.96	28.0	0.98	7.4	0.90
P1	5.4	2.0	0.89	5.5	0.88	0.5 ^{ns}	0.14
P3	4.9	2.0	0.96	5.0	0.96	1.5 ^{ns}	0.71

[†] $p < 0.10$, ^{ns} not significant.

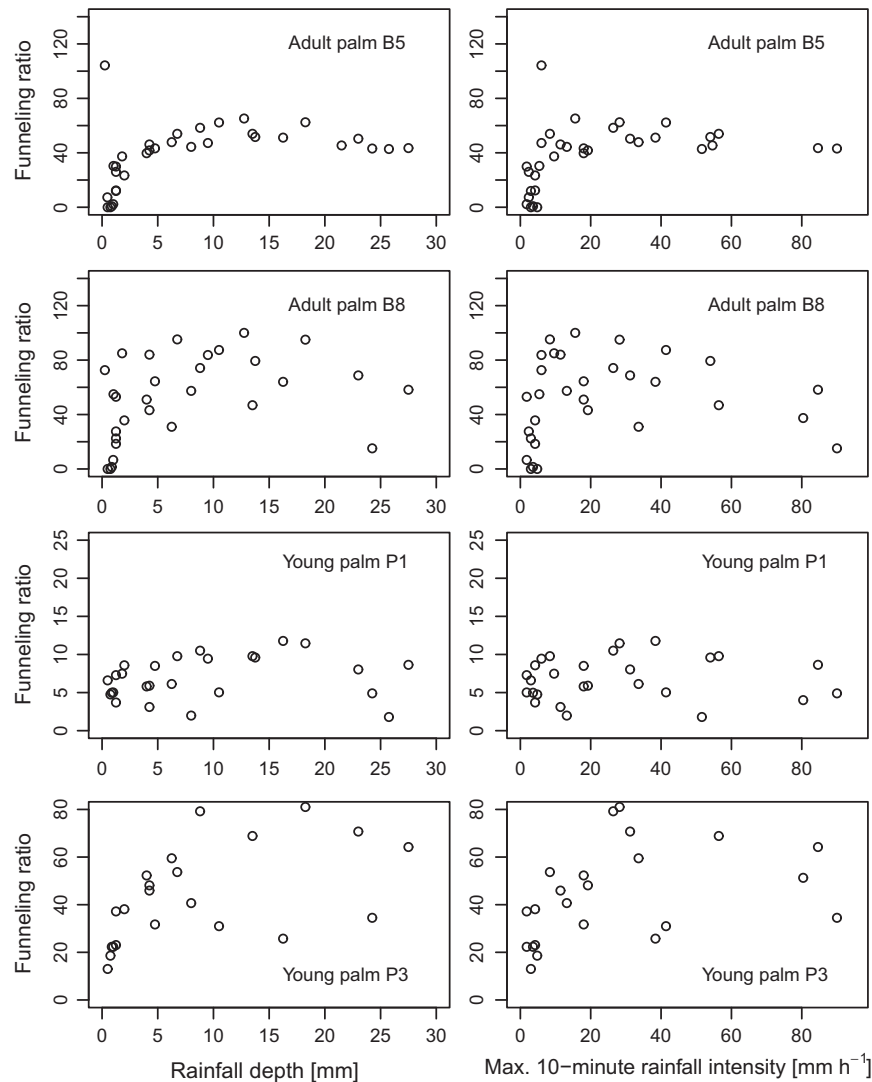


Fig. 4. Funneling ratios per event of two adult and young palms plotted against rainfall depth (mm) and maximum 10-min rainfall intensity (mm h^{-1}).

Table 4

Frequency of perched water table detection at three distances from palm stem and at two soil depths for each distance. For each piezometer (PZ1 to PZ8), the total number of rainfall events during the respective study period is given (n).

Distance (cm)	20		100		200		n
Depth (cm)	20	50	20	50	20	50	
PZ1	22	0	0	0	0	0	40
PZ2	0	1	0	0	2	1	40
PZ3	1	1	0	0	0	0	40
PZ4	0	0	0	0	0	0	40
PZ5	10	1	0	0	–	–	34
PZ6	1	0	1	0	–	–	34
PZ7	8	0	0	0	–	–	34
PZ8	8	0	4	0	–	–	34

(Fig. 7b). As there had only been little rainfall before the tracer experiments on the adult PZ6 and the young palm, the soil around them was already dry and hard. One day after the 10 mm rainfall event simulation, the soil had been moistened up to one meter in depth next to the adult palm stem and up 75 cm horizontal distance away from it (PZ6, Fig. 8). For the respective simulation with the young palm, the wet-dry boundary was 65 cm deep and had a maximal lateral extent of 25 cm (Fig. 7b).

3.4. Root length density distributions

The root length density distributions of the adult palms clearly differed for fine ($<2 \text{ mm}$) and coarse ($\geq 2 \text{ mm}$) roots (Fig. 8). Fine roots were concentrated in the top 10 cm soil layer, and particularly high densities ($>0.4 \text{ cm cm}^{-3}$) were mostly found within 1.5 m from palm stems. Below the top layer, the root density rarely exceeded 0.4 cm cm^{-3} , but roots were distributed rather uniformly over the entire horizontal extent of the profiles and up to a depth of 80 cm. Coarse roots were concentrated next to subterranean palm stems. The radial propagation of coarse roots, which lack secondary growth, resulted in a decrease of coarse-root length density with increasing distance from both subterranean palm stems. The coarse-root length density totals per depth were highest just below the surface layer, at 10–30 cm soil depth.

4. Discussion

4.1. Stemflow generation

For studies reviewed by Levia and Frost (2003), maximum reported stemflow as a percentage of incident rainfall for tropical regions ranged from $<1\%$ to 13.6%. Funnel-like shaped canopies can lead to high stemflow volumes resulting 21–25% stemflow of gross

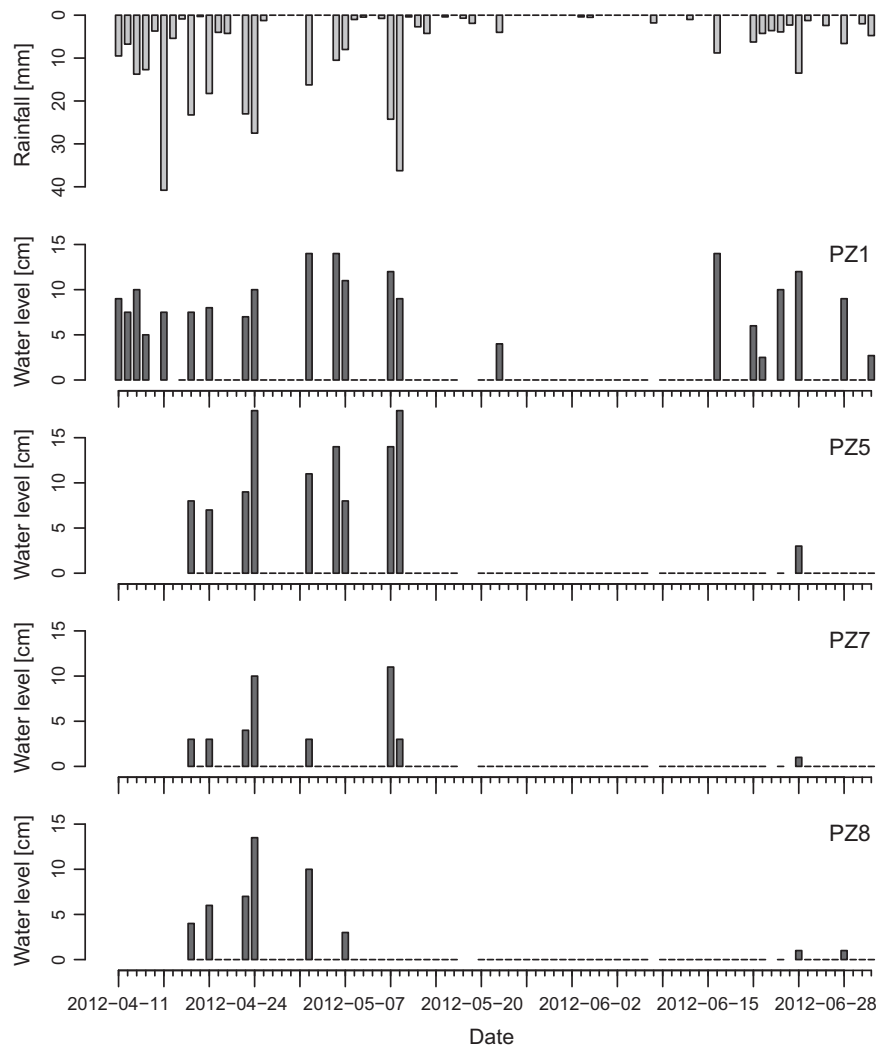


Fig. 5. Rainfall and maximum water levels detected at 20 cm from babassu palm stems in piezometers of 20 cm soil depth.

rainfall in tropical peach palm monocultures with plant densities of 2500 plants per hectare (Schroth et al., 1999). In the same study, however, stemflow of a polyculture including just 234 peach palms per hectare was not more than 3% of incident rainfall. With a median contributing area of $6.4 \pm 0.3 \text{ m}^2$ ($\pm \text{SE}$, value calculated for low intensity events) per palm and a minimum palm density of 156 individuals per hectare (Maranhão State Law No. 4734 from 1986), one-tenth of rainfall will flow down the palm stems. But stemflow funneling ratios have been rarely determined for palms. One exception is a nikau palm (*Rhopalostylis sapida* Wendl. & Drude) in New Zealand, for which Enright (1987) reported a funneling ratio of 142 for his study period. Even though the palm was 5 m tall, it had a DBH of only 16 cm. Exceptionally high funneling ratios, averaging 154 ± 66 , for 44 events were also reported for a semiarid scrub (*Caragana korshinskii* Kom.) with a comparatively small stem diameter (exact value not reported; Li et al., 2008). The adult palm median funneling ratio for the whole study period in the present study was 50 ± 10 , ranging from 16 to 71 for individual palms. This falls in the range of values reported for tropical trees with radii not greater than 15 cm (Germer et al., 2010b; Herwitz, 1986). But it does exceed the funneling ratios of between 14 and 34 (excluding palms with inclined trunks) found for babassu palms within an open rainforest at the Rancho Grande study site in Rondônia, Brazil (Germer et al., 2010b), even though the latter had smaller DBHs (29–33 compared to 29–38 in my study). Palms

in forests are likely to have less space and smaller crown areas for water collection than the isolated palms studied here. Unfortunately, there does not seem to be any existing study comparing stemflow generation of forest trees to that of isolated trees. Nevertheless, my results indicate that there can be substantial differences between such trees and that stemflow funneling values of individual forest trees cannot easily be transferred and used in interception studies in agroforestry systems having much greater distances between trees and without the shading effect of emergent tree canopies.

Median funneling ratios of young (4, 7, and 55) palms were within the range or lower than those of adult palms. This is believed to be partly due to the lower number of palm fronds for young compared to adult palms (Table 2). Other factors might include differences in frond size between young and adult palms as well as lower rainfall input due to the shading effect of neighboring trees.

In the present study, the regression slope of stemflow volume on rainfall depth of individual trees and, hence, the funneling ratios tended to be lower for high- compared to low-intensity rainfall events. This can be attributed to stemflow becoming detached from a frond or trunk, which results in drip points during high-intensity rainfall (personal observation). This effect was previously reported for tropical and temperate forests and desert shrubs (Carrlyle-Moses and Price, 2006; Herwitz, 1987; Levia et al., 2010; Li

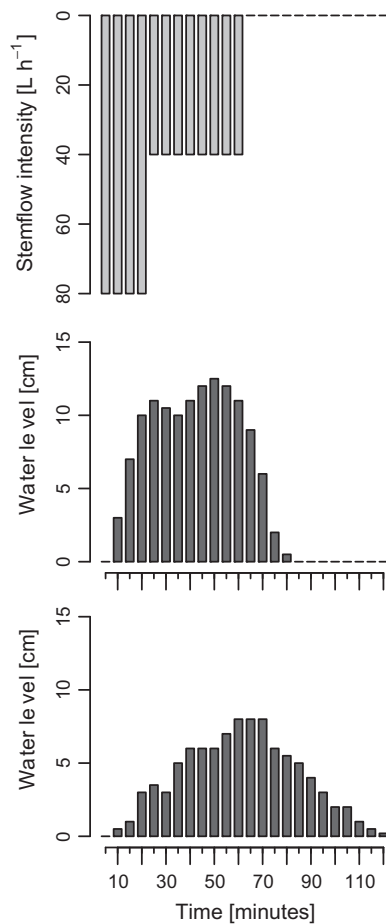


Fig. 6. Stemflow intensity (top plot) and water levels recorded every 5 min at 20 cm from babassu palm stem in a 20 cm deep piezometer (PZ5) on May 15 (middle plot) and May 24 (bottom plot).

et al., 2009). An additional reason for the lower slope of high-intensity rainfall events might be 'splash loss' due to greater raindrop volume and velocity and, hence, kinetic energy during high-intensity events, as suggested by Herwitz (1987) and later by Carlyle-Moses (2004). Yet the same kind of stemflow volume on rainfall depth regression analysis, but using data from babassu palms of

the Rancho Grande forest site resulted in no obvious differences in regression slopes for low- or high-intensity rainfall events (Germer et al., 2010b, data not shown). This might be because, in forest sites, the regression slope is not only affected by high-intensity rainfall events, but also by high interception through nearby tree canopies for low-intensity events. If the canopies of adjacent trees have significant influence on stemflow generation, then it might be appropriate in some cases to choose isolated trees for studying the influence of rainfall characteristics or tree morphology on stemflow.

4.2. Palm trunk storage

Trunk storage (or bark storage) is difficult to quantify, but it can comprise a great share of total canopy storage and is, hence, an important parameter for interception estimation (Crockford and Richardson, 1990; Herwitz, 1985; Reid and Lewis, 2009; Van Dijk and Bruijnzeel, 2001). I could, however, not find any trunk storage values for palms in the literature. For this study, I have calculated average trunk storage per adult babassu palm of 9.1 ± 3.1 L ($\pm$ SE). Based on a minimum density of 156 palms per hectare (Maranhão State Law No. 4734 from 1986) this results in a trunk storage of 0.14 ± 0.05 mm of rainfall. A lower palm density, as found by Gehring et al. (2011), in secondary forests of 120 and 80 individuals per hectare would lead to a trunk storage of 0.10 ± 0.04 mm and 0.07 ± 0.03 mm, respectively. All these values fall within the reported range for trunk storage values of 0.02–0.46 mm for temperate forests or of 0.09–0.22 mm for tropical forests (Germer et al., 2010b; Lloyd et al., 1988; Ubarana, 1996). Therefore, stand density might be at least as important as forest type in interception studies that include stemflow and, hence, trunk storage. The trunk storage of individual babassu palms in a forest canopy compared to isolated palms might, however, evaporate slower due to the shading effect of neighboring trees.

The present study only includes rainfall events during the wet season, with relatively wet antecedent conditions. But, as expected, the trunk storage estimated for events without any rainfall in the previous 24 h was slightly higher with 11.0 ± 3.8 L (SE). This is attributed to the time of at least 24 h that is needed for the trunks to dry. It is likely that the trunk storage is even greater during the dry season. The trunk storage of individual young palms was lower (5.4 and 4.9 L, Table 3) than for adult palms. This was expected as water runs through decomposing organic material that accumulates in the funnel-shaped canopy of adult palms intercepting part of the stemflow. This is, however, not the case for the sampled

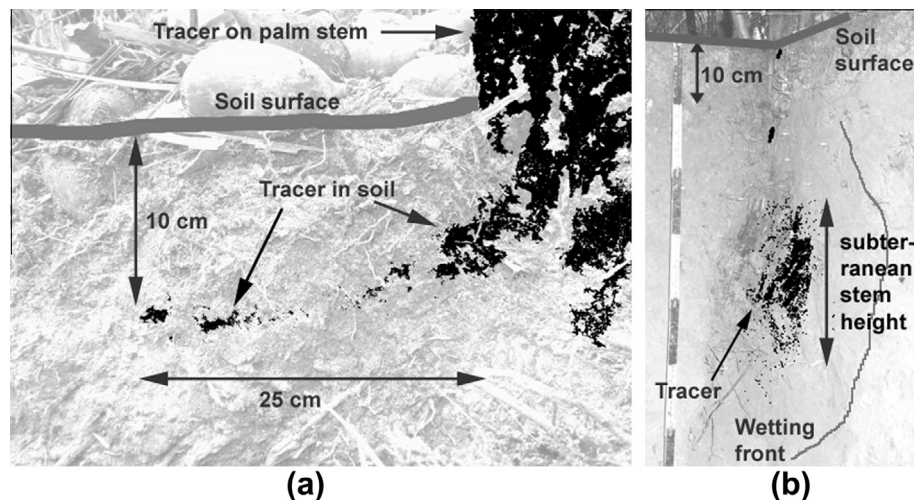


Fig. 7. Stemflow dye tracer experiments from May 25, 2012, showing (a) a horizontal flow pattern for an adult babassu palm (PZ5) and (b) a vertical flow pattern for a young babassu palm (P3).

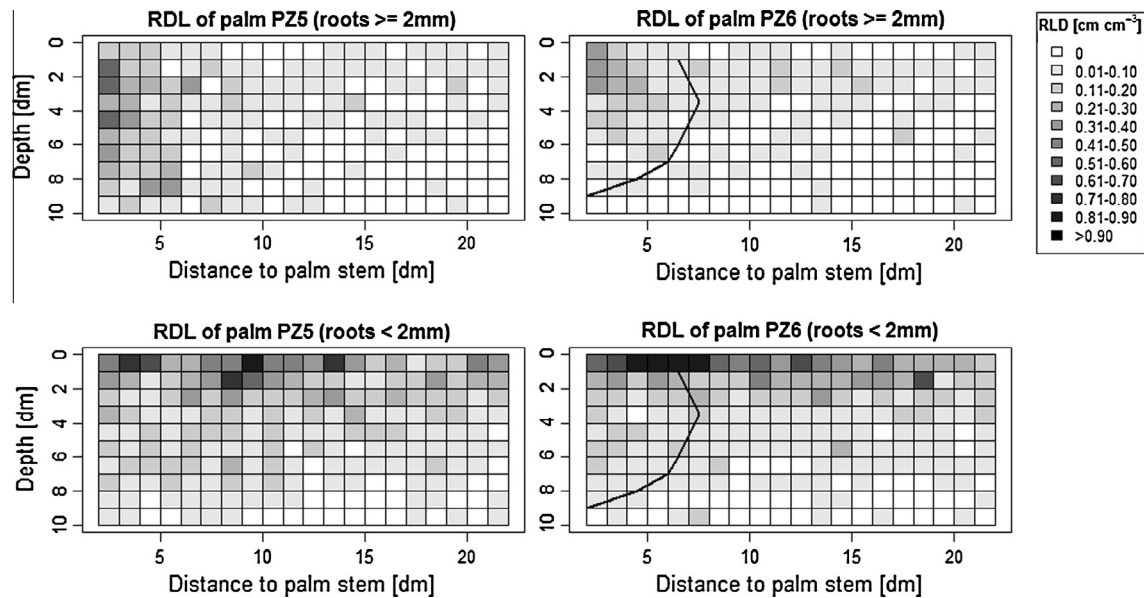


Fig. 8. Coarse and fine root length density (RLD) distributions for two adult babassu palms. For palm PZ6, the extent of the wetting front is shown (black line) as observed one day after the stemflow simulation and tracer experiment.

stemflow from young palms as water was collected directly from the palm fronds.

4.3. Soil–water movement next to palms and their roots

The young babassu palms studied funneled rainfall via their fronds directly to their subterranean stems. For the adult palms, however, water redistribution within the soil was more complex. Perched water tables at 20 cm soil depth were detected more frequently near adult palm stems than further away. Next to some of these palms, perched water tables were never or only rarely observed (Table 4), but it is still possible that saturated conditions existed next to these palms during rainfall events at less than 20 cm soil depth. This was probably the case for palm PZ6, for which the dye tracer experiment provided evidence of lateral preferential flow at 7 cm soil depth, up to 40 cm from the palm stem, for the simulation of a rather small rainfall event (10 mm). As assumed by Herwitz (1986) for stemflow-induced Horton overland flow, the extent of lateral subsurface flow depends on soil hydraulic conductivity as well as on stemflow intensity. Therefore, it can be expected that the extent of a stemflow-impacted soil area not only depends on tree diameter, as discussed by Li et al. (2009), but at least in the case of stemflow-induced soil saturation it also depends on stemflow intensity. This might explain why the lateral preferential flow exhibited a greater extent for the tracer experiment for PZ6 than PZ5, because simulated rainfall intensity was greater for the former. The palm might profit from lateral preferential flow of nutrient rich water as fine roots are concentrated in the topsoil (Fig. 8).

The zone between the perched water table and the lower wetting front is called the transition zone (Ward and Robinson, 2000). As the soil in this zone is only near-saturated, no responses were observed in the 50 cm deep piezometers. Nevertheless, water movement is still much quicker in the transition zone than in soil with lower saturation levels and, hence, leads to ready drainage next to palms where coarse babassu roots are most abundant. The fast drainage might include the downward movement of plant nutrients, which are usually concentrated in stemflow (Levia and Frost, 2003; Parker, 1983). The duration of saturated conditions above the impeding layer and, consequently, of stemflow controls

the maximum depth of the wetting front and, therefore, of quick percolation (Ward and Robinson, 2000). Despite the occurrence of preferential horizontal flow, it can be said that stemflow water infiltrated next to the studied palm stems, as evidenced by the resulting wetting front shape (Fig. 8) and piezometric responses (Table 4). This process is in contrast to the double-funneling of trees concept as only minor root-induced preferential flow was observed in this study (Johnson and Lehmann, 2006; Schwärzel et al., 2012).

The two stemflow simulation events for palm PZ5, having the same stemflow intensities and duration, resulted in interesting differences of piezometric response, with an immediate response to stemflow intensity changes on May 15, but rather delayed intensity response as well as a lower maximum water level on May 24 (Fig. 6). This might be an effect of differences in soil-fauna activity and/or antecedent soil moisture between the dates (rainfall per antecedent week was 66 and 6 mm for May 15 and 24, respectively), which could have led to changes of hydrologic conductivity ratios for soil above and within the impeding layer.

To my knowledge, this is the first study to report direct evidence for stemflow-induced saturation within the upper 20 cm of the soil profile above an impeding layer. High-intensity point water input to the soil surface due to stemflow might, then, lead to three types of response: (a) Horton overland flow (Banabas et al., 2008; Cattani et al., 2007; Herwitz, 1986), (b) subsurface saturation above an impeding layer within the upper 20 cm of the soil profile (this study), or (c) fast, deep percolation as preferential flow along thick roots and subsequent saturated zone generation below the tree at the soil–bedrock interface (Li et al., 2009; Liang et al., 2011).

5. Conclusions

This study showed that, on average, adult babassu palms in a Brazilian agroforestry system collected the amount of rainfall that a collector with an opening of $6.4 \pm 0.3 \text{ m}^2$ ($\pm \text{SE}$) would have collected. In agroforestry systems with 156 palms per hectare that would lead to 10% stemflow of incident rainfall. Whole study period funneling ratios between 16 and 71 for adult or between 4 and 55 for young palms emphasize, however, the high variability of

stemflow generation. Nonetheless, even very small wet-season rainfall events of 1 mm can generate stemflow. For the sampled events, on average 9.1 ± 3.1 L of the rainfall water available for stemflow generation were intercepted by each adult palm trunk (trunk storage). For high-intensity rainfall events, the stemflow contributing area was less effective probably due to splash loss. This led to lower funneling ratios and, hence, less stemflow compared to low-intensity rainfall events of similar size.

The funneling of rainfall towards adult palm stems led to great stemflow fluxes down to the soil and induced initial horizontal water flows through the soil, leading to perched water tables next to palms, even after small rainfall events. The perched water tables extended, however, only a few decimeters from palm stems and, hence, were observed more frequently next to the palms than further away. After perched water tables became established, vertical percolation through the soil dominated. Young babassu palms, in contrast, funnel rainfall via their fronds, directly to their subterranean stems. The root length density distributions differed for fine and coarse roots. Fine roots were concentrated in the topsoil and might profit from preferential horizontal flow of the nutrient rich water at least within an area of a few decimeters from the palm stem. In contrast, coarse root length density was highest next to the subterranean stem and, hence, the palms might profit from increased water input compared to open land rainfall in particular for small rainfall events during the dry season. While other palm species with similar canopy shapes might also funnel large amounts of water towards their stems, flowpath patterns in the soil might, however, be different due to soil textural or structural differences. As oil palms (*Elaeis guineensis* Jacq.) are known to generate high stemflow volumes and due to the widespread cultivation of oil palms in plantations (Banabas et al., 2008), similar studies for this palm about the fate of stemflow water and nutrients after reaching the soil might generate interesting results in relation to water and nutrient management in these ecosystems.

Stemflow of babassu palms can lead to near-surface soil saturation. Soil saturation and simultaneous input of nitrate and presumably dissolved organic carbon could lead to hotspots of denitrification in the soil next to babassu palms. This merits further investigation. Previous studies reviewed by Neill et al. (2005), indicate that N_2O emissions from Amazonian soils are lower from mature pastures compared to forests. As degraded pastures get commonly invaded by babassu palms, potential denitrification hotspots could influence this comparison. In general, the identification of biogeochemical hotspots and research on the underlying processes might help us to evaluate the hotspot utility as tools for resource management (Hellebrand et al., 2008).

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